# 4.1 Divisibility and well-ordering

For integers $a, b$, write $a \mid b$ if $b = ak$ for some integer $k$. Negative divisors are allowed; this relation is not a numerical quotient. For every $m, n \in \mathbb{Z}$,

$$a \mid b,\ a \mid c \Rightarrow a \mid (mb + nc).$$

The well-ordering principle says every nonempty subset of positive integers has a least element. It does not apply directly to every nonempty integer set; all negative integers have no least member. Here $0 \mid b$ holds exactly when $b = 0$, so $0 \mid 0$ is allowed. The reference textbook instead requires the divisor in the divisibility relation to be nonzero. Division algorithms still require a nonzero divisor.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

## 4.2 The precise form of weak induction

To prove $\forall n \geq n_0, P(n)$, prove $P(n_0)$, then for arbitrary fixed $k \geq n_0$, assume $P(k)$ and derive $P(k+1)$. This covers the successor chain starting at $n_0$. The base is essential; a skipped or incorrect starting point may invalidate the claim.

For the false claim "all integers $n \geq 2$ are even," moving from $2k$ to $2(k+1)$ covers only even integers. It does not prove the required implication for an arbitrary index. The step may use $P(k)$ repeatedly or split into cases according to $k$, while keeping $k$ arbitrary and fixed.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.2.1 Induction follows from well-ordering

Induction is not confidence gained by checking many successors. Suppose $P(n_0)$ and $P(k) \to P(k+1)$ for every $k \geq n_0$. Let

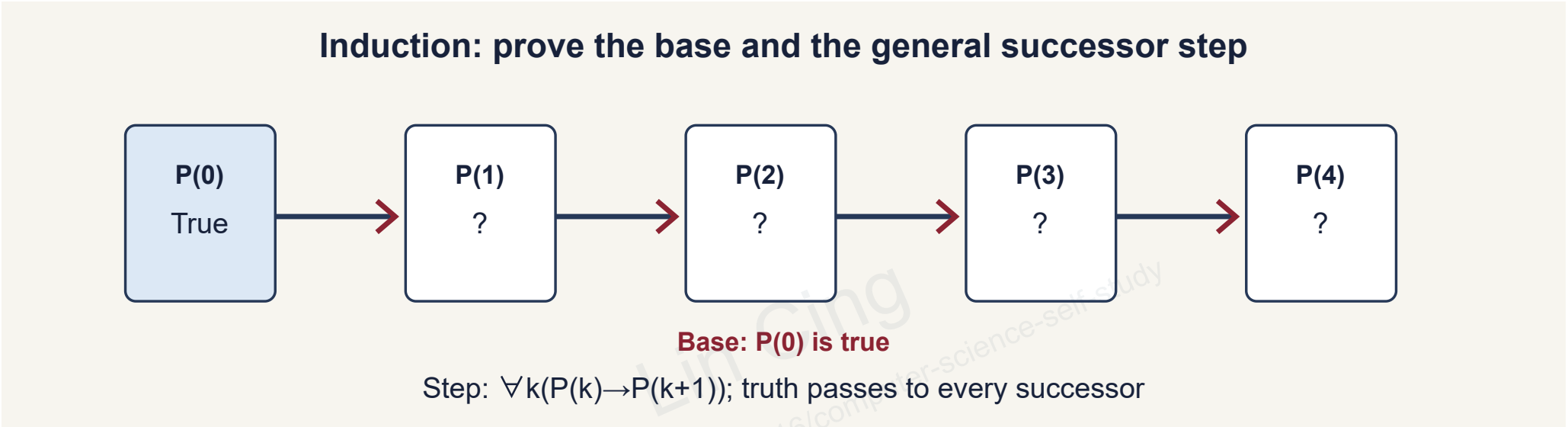
$$S = \{n \in \mathbb{Z} : n \geq n_0 \wedge \neg P(n)\}.$$

If the theorem fails, $S$ is nonempty. Its translate $\{n - n_0 + 1 : n \in S\}$ is a nonempty positive-integer set, so well-ordering gives a least counterexample $m$. The true base excludes $m = n_0$. Therefore $m - 1 \geq n_0$, and minimality gives $P(m-1)$. The induction step gives $P(m)$, contradicting $m \in S$. Thus $S$ is empty.

Without a base, this argument cannot exclude failure at $n_0$. Without the step, it cannot pass to the successor.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# The induction hypothesis and domino model




Assuming $P(k)$ is a conditional assumption for one arbitrary fixed index, not an assumption of the whole theorem. The completed step proves an implication.

# 4.2.2 Complete induction examples and boundaries

For $n \geq 1$, prove

$$1 + 3 + \cdots + (2n - 1) = n^2.$$

At $n = 1$, both sides equal one. Assume the formula at fixed $k \geq 1$. The next term is $2(k + 1) - 1 = 2k + 1$, and

$$1 + 3 + \cdots + (2k - 1) + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2.$$

This is the next-index claim, completing induction.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Induction with a later starting point

For $P(n) : 4n < n^2 - 7, n \geq 6$, the base gives $24 < 29$. Assume $4k < k^2 - 7$ with $k \geq 6$. Then

$$4(k+1) = 4k + 4 < k^2 - 3 < k^2 + 2k - 6 = (k+1)^2 - 7,$$

where the second inequality uses $2k > 3$. The claim fails at one, since $4 < -6$ is false, so its start cannot be changed to one.

Writing $P(k+1)$ on scratch paper helps find a route from $P(k)$. The formal proof must start with established assumptions and end at the target, without assuming the target.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.3 Strong induction and least counterexamples

To prove $P(n)$, strong induction assumes every earlier $P(j)$, $n_0 \leq j < n$. It is useful for integer factorization, graph-removal operations, and divide-and-conquer structures. Weak induction is a special case.

A least-counterexample proof forms $S = \{n \geq n_0 : \neg P(n)\}$, assumes it nonempty, and applies well-ordering to derive a smaller counterexample or contradiction. This method differs from contraposition and must explicitly justify nonemptiness.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.3.1 Equivalence of weak and strong induction

Assume weak induction valid. Suppose $P(n_0)$ holds and all preceding statements imply $P(n)$ for each $n > n_0$. Set

$$Q(n) = P(n_0) \land P(n_0 + 1) \land \cdots \land P(n).$$

The base $Q(n_0)$ holds. Assuming $Q(k)$ supplies every earlier $P$; the given strong step yields $P(k + 1)$, hence $Q(k + 1)$. Weak induction proves every $Q(n)$, and $Q(n) \Rightarrow P(n)$ proves the desired result. Conversely, strong induction includes the most recent hypothesis $P(k)$, so it permits every weak-induction proof. It broadens the usable history without increasing logical proof strength.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Strong induction for a two-term recurrence

For a complete example , let $a_0 = 1, a_1 = 2$, and

$$a_n = 2a_{n-1} + 3a_{n-2} \quad (n \geq 2).$$

Claim $a_n = (3^{n+1} + (-1)^n)/4$. At zero and one, the formula gives $(3 + 1)/4 = 1$ and $(9 - 1)/4 = 2$. Fix $k \geq 1$ and assume the formula through $k$. Then

$$\begin{aligned} a_{k+1} &= 2a_k + 3a_{k-1} \\ &= \frac{6 \cdot 3^k + 3 \cdot 3^k + 2(-1)^k - 3(-1)^k}{4} \\ &= \frac{9 \cdot 3^k - (-1)^k}{4} = \frac{3^{k+2} + (-1)^{k+1}}{4}. \end{aligned}$$

The two-term recurrence needs two bases to start the dependence on $k - 1$. Every strong proof can be rewritten as weak induction on a prefix conjunction.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

# Strong induction: stamps of values 3 and 8

## Three residue chains: add 3 at each step

Base: 14 = 3 + 3 + 8




Base: 15 = 5 × 3




Base: 16 = 8 + 8




We claim every integer $n \geq 14$ can be formed using stamps worth 3 and 8. The base cases are $14 = 3 + 3 + 8$, $15 = 5 \cdot 3$, and $16 = 8 + 8$. Adding 3 preserves feasibility; the three bases cover the three residues.

# Stamp example: the inductive step

Fix an integer $n \geq 17$, and assume every value $14, 15, \ldots, n-1$ can be formed. Since $14 \leq n-3 < n$, the hypothesis applies to $n-3$. Add one 3-value stamp to obtain $n$.

The bases cover $14, 15, 16$, and the step covers every $n \geq 17$. Checking only 14 would not start the other two residue chains.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.4 Recursive definitions

A recursive definition specifies bases and a rule, such as

$$F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2} \quad (n \geq 2).$$

Existence and uniqueness both need justification. If every new value depends only on smaller indices, finite iteration determines it uniquely. Negative-index references or insufficient initial values can leave a sequence undefined.

More generally, let $r \geq 1$ be an integer, let $S$ be a nonempty set of values, and choose bases $a_0, \ldots, a_{r-1} \in S$. Require $f$ to be defined on every pair of an integer $n \geq r$ and an $S$-valued prefix of length $n$, with values in $S$. Then

$$a_n = f(n, a_0, \ldots, a_{n-1}) \quad (n \geq r)$$

defines exactly one sequence. The recursion theorem constructs the infinite sequence: each finite prefix extends uniquely by the total function $f$, and these prefixes are compatible, so their entries determine one sequence. If another sequence has the same bases and rule, induction on prefix length shows all its entries agree; hence it is unique.

# Recursive initial values and uniqueness

For Fibonacci bases $b_0 = 0, b_1 = 1$, iteration gives $b_2 = 1, b_3 = 2, b_4 = 3, b_5 = 5, b_6 = 8, b_7 = 13$. Without the two bases, both $0, 1, 1, 2, \ldots$ and $2, 5, 7, 12, \ldots$ satisfy the same recurrence. Nor does $d_n = d_{n+1}$ without other boundaries uniquely determine an infinite sequence. Recursion describes a definition method; uniqueness depends on well-founded structure and initial conditions.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Harmonic and Lucas sequences: starting values

Set $H_0 = 0$ and, for integers $n \geq 1$, define $H_n = H_{n-1} + 1/n$. Successive values are:

| $n$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $H_n$ | 0 | 1 | $3/2$ | $11/6$ | $25/12$ |

For example, $H_4 = H_3 + 1/4 = 11/6 + 1/4 = 25/12$. The value $H_0 = 0$ is the empty-sum convention; it does not introduce a $1/0$ term.

The Lucas sequence starts at $L_0 = 2, L_1 = 1$ and satisfies $L_n = L_{n-1} + L_{n-2}$ for $n \geq 2$, giving $L_2 = 3, L_3 = 4, L_4 = 7$. Its recurrence matches Fibonacci's, but its starting values differ, so the recurrence alone does not make the sequences equal.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Eulerian numbers: classify permutations by descents

For integers $n \geq 1$, let $A(n, k)$ count permutations of $1, \ldots, n$ with exactly $k$ descents, where a descent is an adjacent position with $\pi_i > \pi_{i+1}$. Use $A(0, 0) = 1$; for $n \geq 1$, set $A(n, k) = 0$ when $k < 0$ or $k \geq n$, and set $A(0, k) = 0$ for $k \neq 0$. Thus $A(1, 0) = 1$.

For $n \geq 2$, insert the largest element $n$, separating the cases by the old descent count:

- A permutation with $k$ descents has $k + 1$ preserving positions: inside any descent or at the end.
- A permutation with $k - 1$ descents has $n - k$ increasing positions: at the start or inside any ascent.

Deleting the largest element uniquely recovers the old permutation and insertion position. The cases are disjoint and exhaustive, giving

$$A(n, k) = (k + 1)A(n - 1, k) + (n - k)A(n - 1, k - 1).$$

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Eulerian recurrence on three elements

Using $A(2,0) = A(2,1) = 1$ and the out-of-range convention:

$$A(3,0) = 1, \quad A(3,1) = 2(1) + 2(1) = 4, \quad A(3,2) = 1.$$

| Permutation | Descents | Permutation | Descents |
|---|---|---|---|
| 123 | 0 | 231 | 1 |
| 132 | 1 | 312 | 1 |
| 213 | 1 | 321 | 2 |

Hence $1 + 4 + 1 = 6 = 3!$. In general every permutation has exactly one descent count, so the row sum is $n!$ by a disjoint classification; it is not a conclusion based only on the first three rows.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.5 Recursive algorithms and structural induction

Recursive factorial needs an explicit base return and calls with strictly smaller arguments remaining inside the domain. Correctness can be proved by induction on argument size.

For $\mathrm{bitsum}(0) = 0$, define for $n > 0$,

$$\mathrm{bitsum}(n) = \mathrm{bitsum}(\lfloor n/2 \rfloor) + (n \bmod 2).$$

The nonnegative argument strictly decreases, so recursion terminates. A trace is

$$\mathrm{bitsum}(45) = 1 + \mathrm{bitsum}(22) = 1 + 0 + \mathrm{bitsum}(11) = \cdots = 4.$$

The full remainder sequence is $1, 0, 1, 1, 0, 1$, matching four ones in $45 = (101101)_2$. For correctness, strong induction assumes all smaller arguments give their binary digit sums, then adds the last binary digit in the displayed recurrence. Termination and correctness are separate claims. Recursion trees or the master theorem analyze complexity.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.6 The division algorithm

For integer $a$ and positive integer $d$, unique integers $q, r$ satisfy

$$a = qd + r, \qquad 0 \leq r < d.$$

For existence, choose $q = \lfloor a/d \rfloor, r = a - qd$. For uniqueness, two representations give $r - r' = (q' - q)d$, while $-d < r - r' < d$, forcing zero.

Programming-language remainder rules for negative inputs can differ. In particular C/C++ `%` follows its quotient convention rather than the nonnegative mathematical convention for $a \bmod d$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

## 4.6.1 A well-ordering proof of division

The floor construction is short; well-ordering explains existence of a least nonnegative remainder directly. For fixed $a \in \mathbb{Z}, d > 0$, form

$$S = \{a - dq : q \in \mathbb{Z}, a - dq \geq 0\}.$$

It is nonempty: $q_0 = -|a| - 1$ gives $a + d(|a| + 1) > 0$, since $d \geq 1$. Let its least member be $r = a - dq$. If $r \geq d$, then $r - d = a - d(q + 1)$ is a smaller nonnegative member, contradiction. Thus $0 \leq r < d$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Uniqueness of quotient and remainder

For another representation $a = dq' + r'$ in the same range,

$$r - r' = d(q' - q), \qquad -d < r - r' < d.$$

Only zero in that interval is divisible by $d$, so $r = r'$. Since $d > 0$, also $q = q'$.

**Mathematical division**

**-37 = 8(-5) + 3**

q=-5，r=3

Condition: 0 ≤ r < 8

The mathematical quotient and remainder are unique

**Truncation and nonnegative remainders**

**-37 = 8(-4) + (-5)**

C/C++: -37 % 8 is -5

For 0 ≤ r, adjust r += 8 and q -= 1

Program rules can differ from the mathematical convention

**Specify quotient and remainder conventions before evaluating code behavior**

Exhibiting one pair proves existence but not uniqueness. Excluding a second pair proves uniqueness but not existence.

# 4.6.2 Base representations and two's complement

For nonnegative integer $n$, repeated division in integer base $b \geq 2$ gives

$$n = r_0 + r_1 b + \cdots + r_k b^k, \qquad 0 \leq r_i < b.$$

For example,

$$47 = 3 \cdot 15 + 2, \quad 15 = 3 \cdot 5 + 0, \quad 5 = 3 \cdot 1 + 2, \quad 1 = 3 \cdot 0 + 1.$$

Reading remainders upward gives $47 = (1202)_3$, checked by $27 + 18 + 0 + 2 = 47$.

For positive input, each quotient $\lfloor n/b \rfloor < n$, so division terminates at zero. Substitution back proves existence with a nonzero leading digit. Any two expansions have equal last digits modulo $b$. Subtract that digit, divide by $b$, and repeat to prove uniqueness. Write zero as one digit 0. Negative integers use a separate minus sign rather than a sum of nonnegative digits times positive powers.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Two's complement and overflow

For eight-bit two's complement , write $6 = (00000110)_2$, invert bits to $11111001$, and add one:

$$-6 = (11111010)_2.$$

Precisely, $w$ bits represent

$$-b_{w-1}2^{w-1} + \sum_{i=0}^{w-2} b_i 2^i,$$

with range $[-2^{w-1}, 2^{w-1} - 1]$. When the top bit is one, subtract $2^w$ from unsigned value $U$. For minus six in eight bits, $U = 256 - 6 = 250$. The top bit has negative weight, rather than being an attached sign label.

Inverting and adding one again recovers $00000110$. Fixed-width signed addition may discard the left carry, yet equal-sign operands producing the opposite sign indicate overflow. Eight-bit $127 + 1$ produces $10000000$, interpreted as minus 128. Bit patterns and mathematical integer values must be distinguished. Base representation proceeds low digit first, and two's complement requires a fixed width and range convention.

# 4.7 Primes and irreducibles require full definitions

An integer $p > 1$ is prime when its only positive divisors are one and itself. One is a unit, not prime; zero is not prime either.

In an integral domain $R$, a prime element $p$ must be nonzero and a nonunit, and satisfy

$$p \mid ab \Rightarrow p \mid a \text{ or } p \mid b \qquad (a, b \in R).$$

A unit has a multiplicative inverse. An irreducible is also nonzero and a nonunit, and every factorization $p = ab$ has a unit factor.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Prime element conditions and examples

| Element | Can the divisibility implication alone hold? | Prime element? |
|---|---|---|
| 0 | Yes in a domain, since $ab = 0$ forces a zero factor | No, nonzero condition fails |
| Units such as $\pm 1$ in integers | Yes, units divide everything | No, nonunit condition fails |
| 2 in integers | Yes, by Euclid's lemma | Yes, and irreducible |
| 4 in integers | No, it divides $2 \cdot 2$ but neither factor | No; reducible as $2 \cdot 2$ |

A prime is irreducible: if $p = ab$ and $p \mid a$, put $a = pc$ and cancel nonzero $p$ to get $1 = cb$, making $b$ a unit; the other case is symmetric. The converse holds in integers but may fail in other domains. For trial division of positive $n$, a factorization $n = ab, 1 < a \leq b$ gives $a \leq \sqrt{n}$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.7.1 Prime divisors and infinitely many primes

For composite $n > 1$, its nontrivial divisors $1 < d < n$ form a nonempty set. Let $p$ be the least. If $p = ab$ with $1 < a < p$, then $a$ is a smaller nontrivial divisor of $n$, contradiction. Thus $p$ is prime. Writing $n = pk$, minimality gives $p \leq k$, so $p^2 \leq n$. Every composite has a prime divisor at most $\sqrt{n}$, explaining the trial-division stopping condition.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Infinitely many primes

If all primes were $p_1, \ldots, p_t$, let

$$N = p_1 \cdots p_t + 1.$$

Since $N > 1$, either $N$ is prime and is its own prime divisor, or it is composite and the preceding argument gives a prime divisor. Every listed prime leaves remainder one on dividing $N$, so none divides it, contradiction. Thus infinitely many primes exist.

Allowing one as prime would destroy uniqueness through arbitrary factors of one. Zero is also excluded. For $n = 2$, no trial candidates $2 \leq d \leq \sqrt{n}$ exist, correctly declaring it prime. The test assumes $n > 1$ and does not apply to one.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.8 GCD and the Euclidean algorithm

For integers $a, b$ not both zero, their gcd is the greatest positive common divisor. For $b > 0$,

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

If $a = qb + r$, the common divisors of $a, b$ are exactly those of $b, r$. Successive division reduces remainders to zero, leaving the previous positive remainder as gcd.

A zero input returns the absolute value of the other in constant steps. For two nonzero inputs normalized positive, remainders at least halve every two steps. This gives $O(\log \min(|a|, |b|) + 1)$ divisions, not the bit complexity of large-integer arithmetic.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.8.1 The least positive linear combination

For $a, b$ not both zero, form

$$S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\}.$$

If $a \neq 0$, one of $a, -a$ is positive; if $a = 0$, one of $b, -b$ is positive. Thus $S$ is nonempty. Let its least member be $d = ax_0 + by_0$.

Divide $a = dq + r$, $0 \leq r < d$. Then

$$r = a - dq = a - (ax_0 + by_0)q = a(1 - x_0 q) + b(-y_0 q).$$

If $r > 0$, this is a smaller positive member of $S$, contradiction. Hence $d \mid a$, and similarly $d \mid b$. Every common divisor $c$ divides $ax_0 + by_0 = d$; every positive common divisor is therefore at most $d$. This proves $d = \gcd(a, b)$ and existence of a Bézout representation.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Euclidean invariants and substitution

For Euclidean correctness ,

$$d \mid a \text{ and } d \mid b \iff d \mid b \text{ and } d \mid r$$

follows in the two directions from $r = a - qb$ and $a = qb + r$. The common-divisor set is invariant. The final nonzero remainder is its gcd, and strict remainder descent proves termination.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Euclidean invariants and substitution (continued)

## Euclidean algorithm and Bézout substitution

### Division steps

1. $1071 = 2 \times 462 + 147$
2. $462 = 3 \times 147 + 21$
3. $147 = 7 \times 21 + 0$

Remainders decrease: $147 \to 21 \to 0$

### Common divisors stay unchanged

$C(a,b)$ = set of positive common divisors

$C(1071, 462) = C(462, 147)$

$= C(147, 21) = C(21, 0)$

$r = a - qb$ gives the same common divisors

Remainder 0: last nonzero remainder = gcd = 21

Back-substitution: replace 147 with $1071 - 2 \times 462$

$$21 = 462 - 3 \times 147 \to 21 = 7 \times 462 - 3 \times 1071$$

Coefficients: $1071 \times (-3) + 462 \times 7 = 21$

Calling $d$ the least positive combination does not alone prove maximality. One must show both that it is a common divisor and that all common divisors divide it.

# 4.9 Bézout and linear Diophantine equations

For $a, b$ not both zero, there are integers $s, t$ with

$$\gcd(a, b) = sa + tb.$$

Extended Euclid retains these coefficients; reversing its division steps constructs them. They may be negative and are not unique.

For $84 = 2 \cdot 30 + 24$, $30 = 24 + 6$,

$$6 = 30 - 24 = 30 - (84 - 2 \cdot 30) = 3 \cdot 30 - 84.$$

Thus $84x + 30y = 12$ is solvable, since six divides twelve. Multiply by two to obtain $-2 \cdot 84 + 6 \cdot 30 = 12$, with $(x, y) = (-2, 6)$. If $d = sa + tb$ and $g = \gcd(a, b)$, then for every integer $k$,

$$d = (s + k\,b/g)a + (t - k\,a/g)b.$$

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Solvability of linear Diophantine equations

More generally, for $a, b$ not both zero,

$$ax + by = c$$

has integer solutions iff $g = \gcd(a, b)$ divides $c$. If $c = g\ell$, multiply $g = sa + tb$ by $\ell$ to construct a solution. Conversely, every common divisor divides the left side of any solution, hence divides $c$.

Handle $a = b = 0$ separately: every integer pair solves it when $c = 0$, and none when $c \neq 0$. This convention leaves $\gcd(0, 0)$ undefined. Chapter 14 uses Bézout to test modular invertibility.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

## 4.9.1 GCD, LCM, and common multiples

For positive integers $a, b$, the lcm is their least positive common multiple. We prove

$$\gcd(a,b)\,\operatorname{lcm}(a,b) = ab.$$

Write $a = gA, b = gB$, where $g = \gcd(a,b)$ and $\gcd(A,B) = 1$. The candidate $L = gAB = aB = bA$ is a common multiple.

Any positive common multiple has $M = au = bv$, giving $Au = Bv$. Bézout gives $sA + tB = 1$. Multiply by $v$: both terms on the left are divisible by $A$, the second because $Bv = Au$. Thus $A \mid v$; write $v = Aw$. Then $M = gB(Aw) = Lw$, so $L$ divides every positive common multiple and is least. Finally $gL = g^2AB = ab$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# LCM example and domain

For $\gcd(168, 456) = 24$,

$$\operatorname{lcm}(168, 456) = 168 \cdot 456/24 = 3192.$$

Indeed $168 = 24 \cdot 7, 456 = 24 \cdot 19$, with coprime 7 and 19. Every common multiple must include their product after removing the common factor.

The displayed identity uses positive inputs. Other integer, sign, and zero conventions must be specified first. For example $\gcd(0, n) = |n|$, while $\operatorname{lcm}(0, n)$ requires its own convention.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

## 4.9.2 Prime and irreducible can differ

In a commutative integral domain with identity, both definitions require a nonzero nonunit. Prime means $p \mid ab \Rightarrow p \mid a$ or $p \mid b$; irreducible means $p = ab \Rightarrow$ at least one factor is a unit.

For prime $p = ab$, primality gives $p \mid a$ or $p \mid b$. In the first case $a = pc$ gives $p = pbc$; cancel nonzero $p$ in the domain to obtain $1 = bc$, so $b$ is a unit. The other case is symmetric.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Irreducible integers are prime

In integers the converse holds. For any irreducible integer $p$, let $q = |p| > 1$. The numbers differ by the unit $\pm 1$, so $q$ is irreducible and $p \mid z$ iff $q \mid z$. It therefore suffices to prove primality for a positive irreducible; write it as $p > 1$. For $p \mid ab, p \nmid a$, its only positive divisors imply $\gcd(p, a) = 1$. Bézout gives $up + va = 1$. Multiply by $b$: $upb + vab = b$. Both terms are divisible by $p$, giving $p \mid b$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# A counterexample in a general integral domain

The converse fails in

$$R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}.$$

Define $N(a + b\sqrt{-5}) = a^2 + 5b^2$. Multiplication expands to give $N(uv) = N(u)N(v)$; nonzero elements have positive integer norms. Norm one occurs exactly at $\pm 1$, precisely the units, since inverse norms must multiply to one.

If $3 = uv$ with both factors nonunits, their norms multiply to nine and must both equal three. But $a^2 + 5b^2 = 3$ has no integer solution: nonzero $b$ gives at least five, and $b = 0$ requires $a^2 = 3$. Thus three is irreducible. It is false that all nonzero norms below nine are units; two has norm four and $\sqrt{-5}$ norm five.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

# Why three is not prime in this domain

In the same domain,

$$(2 + \sqrt{-5})(2 - \sqrt{-5}) = 9$$

is divisible by three, while neither factor is: comparing integer coefficients would require three to divide two and $\pm 1$. Three is not prime in this domain. The integer equivalence cannot be transferred to arbitrary domains. The prime divisibility implication alone admits zero and units, so the nonzero and nonunit conditions are essential.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# 4.10 The fundamental theorem of arithmetic

Every integer $n \geq 2$ is a product of primes, uniquely up to order. For existence, the base two is already prime. Assume all integers from two to $n - 1$ factor. If $n$ is prime, stop. Otherwise $n = ab$ with $1 < a, b < n$. Strong induction factors both, and concatenation factors $n$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Euclid's lemma and uniqueness

For uniqueness, first prove Euclid's lemma without using unique factorization. If prime $p \mid ab$ and $p \nmid a$, then $\gcd(p, a) = 1$, so Bézout gives $up + va = 1$. Multiplication by $b$ gives $upb + vab = b$. Both left terms are divisible by $p$, so $p \mid b$.

Extend the lemma to any finite product by induction on its length. The one-factor case is immediate. If $p \mid (a_1 \cdots a_m)a_{m+1}$, the lemma gives divisibility of the last factor or the first product. In the latter case, the induction hypothesis supplies a divisible factor among the first $m$.

Now suppose

$$n = p_1 \cdots p_r = q_1 \cdots q_s$$

are prime factorizations. The extended lemma makes $p_1$ divide some $q_j$, so the two positive primes are equal. Reorder and cancel them, then repeat. If one list ends before the other, a remaining product of primes greater than one would equal one, impossible. Thus the prime lists match with multiplicities, proving uniqueness up to order. In a unique factorization domain, factorizations are unique up to order and multiplication by units; a general integral domain need not have unique factorization.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

# 4.10.1 Standard form, divisor counts, and squares

Write the unique factor multiplicities as

$$n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}, \qquad \alpha_i \geq 1.$$

Every positive divisor uniquely has $d = p_1^{\beta_1} \cdots p_r^{\beta_r}$, $0 \leq \beta_i \leq \alpha_i$. Independent exponent choices and uniqueness give

$$\tau(n) = \prod_i (\alpha_i + 1).$$

A divisor is a square iff all its exponents are even, so square divisors number

$$\prod_i \left( \left\lfloor \frac{\alpha_i}{2} \right\rfloor + 1 \right).$$

For $360 = 2^3 \cdot 3^2 \cdot 5$, $\tau(360) = 4 \cdot 3 \cdot 2 = 24$. Square exponents for two are 0,2; for three 0,2; and for five only zero. The four square divisors are $1, 4, 9, 36$.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

# Square divisors and odd divisor counts

If $n = m^2$, every prime exponent is even; conversely halving even exponents constructs a square root. Also $\tau(n)$ is odd iff every $\alpha_i + 1$ is odd, equivalently every $\alpha_i$ even, so iff $n$ is a square.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

# Trailing zeros of 100!: count factors by level

## Counting factors of 5 in 100!

| | | |
|---|---|---|
| $5^1$ | $\lfloor 100 / 5 \rfloor = 20$ | 5, 10, …, 100 |
| $5^2$ | $\lfloor 100 / 25 \rfloor = 4$ | 25, 50, 75, 100 |
| $5^3$ | $\lfloor 100 / 125 \rfloor = 0$ | 125 > 100 |

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Each multiple of 5 contributes a factor of 5; each multiple of 25 contributes one more. Thus $v_5(100!) = 20 + 4 + 0 = 24$. The 50 even factors supply at least 50 factors of 2, enough to pair with all 24 factors of 5, so there are **24 trailing zeros**.

## 4.11 Proof defects and boundaries

"Every $n \geq 2$ is a product of two integers greater than one" is false at every prime. The corrected conditional for nonprime $n$ needs a precise composite definition.

A strong-induction factorization proof cannot invoke $P(1)$ before establishing its actual base $n = 2$. Every recursive reference must lie in an established or strictly smaller part of the domain.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

## 4.12 Checks and exercises

Check the quantified starting point, bases, decreasing measure, uniqueness of division, gcd invariant, and assumptions for prime-factorization uniqueness. The separate exercise PDF provides questions, and the solution PDF gives their derivations.

### References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 4.

Original educational synthesis by Lin Cing, licensed under CC BY 4.0.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study